

Programming Fundamentals – Lab sheet 07

Question 01) – student performance analyzer

```
total = 0
avg = 0
c_f_sub = 0

for i in range (1,7,1):
    mark = float(input(f"Enter mark {i}:"))
    #total = total + mark
    total+=mark
    #print(total)

    if mark >50:
        c_f_sub = c_f_sub + 1
        c_f_sub+=1

avg = total/6

print("Total marks:", total)
print("Average:", avg)
print("Failed subjects:", c_f_sub)

if avg >= 60 and c_f_sub == 0:
    print("Pass")
else:
    print("Fail")
```

Question 02) ATM system with limited attempts and balance checker

```
balance = 0
attempt = 0
correct_pin = "1234"
print(balance)

while attempt <=3:
    pin = input("Enter your PIN:")

    if pin == correct_pin:
        print("PIN Accepted")

        amount=float (input("Enter the amount:"))

        if amount > balance:
            print("Less balance.")

            elif balance - amount < 500:
                print("Remain should be 500.")
            else:
                #balance = balance - amount
                balance-= amount
                print("retrival is done remain is: ", balance)

                break:
            else:
                attempt +=1
                print("PIN not accepted.")

    if attempt == 3:
        print("No attempts left.")
```

Question 03) Supermarket Billing system

```
price= float(input("Enter the price of the item:"))
total = 0
discount = 0

if price > 10000:
    discount = price * 0.10
    print("Total is: ",total)
    print("discount value is:", discount)
    price("Final total is:", total - discount)
elif price > 5000:
    discount = price * 0.15
    print("Total is: ",total)
    print("discount value is:", discount)
    price("Final total is:", total - discount)
else price < 5000:
    print("No discount is available")
    print("Final total is:", total)
```

Question 04) Bank transaction monitor

```
balance = 0
while True:
    amount = float(input("enter transaction amount deposit +/- withdrawal amount: "))
    if amount == 0:
        break

    balance += amount

print("Final balance: Rs.",balance)
if balance < 0:
    print("Account status: Overdrawn")
else:
    print("account status: Active")
```

Question 05) Electricity bill calculator with slab rates

```
total_units = 0
for i in range(1,5):
    units = float(input(f"Enter the number of units for month {i}: "))
    total_units += units
if total_units <= 100:
    bill = total_units * 10
elif total_units <= 200:
    bill = (100*10) + ((total_units - 100) * 15)
else:
    bill = (100*10) + (100*15) + ((total_units - 200) * 20)

print("Total units used:", total_units)
print("Total bill is: Rs.",bill)
if total_units > 400:
    print("Usage category : High")
else:
    print("Usage category : Normal")
```